

PROJECT SENTINEL-BLR

Evidence-Quality Automated Traffic Enforcement Framework

Submission Details	
Hackathon	Flipkart Gridlock 2.0
Phase	Phase 2
Theme	Theme 3 — Automated Photo Identification and Classification for Traffic Violations
Submission Type	Framework & Prototype Proposal
Version	v4.2 — Adaptive Temporal Striding, Hybrid Evidence Payloads, Volumetric BB Profiling, Mathematical Calibration Grounding

1. Executive Summary

Sentinel-BLR is not a traffic violation detector. It is an evidence-quality system — one that determines when a detected violation is legally reliable enough for enforcement.

Most submissions for this problem will bolt together YOLO, OCR, and a violation classifier. Sentinel-BLR does that too — and then solves the three problems that make those pipelines fail in actual Bengaluru deployment:

- **Cameras that drift** — addressed by a self-healing Auto-Calibration Heartbeat (SIFT/ORB + inverse homography H^{-1}) detecting and correcting spatial drift every 15 minutes, without halting enforcement.
- **Plates that are unreadable** — addressed by Temporal Character-Level Probability Fusion across 5 frames with Levenshtein outlier filtering, producing a majority-vote plate string even from mud-damaged plates.
- **Detection confidence that does not translate into legally defensible enforcement** — addressed by per-violation-type confidence thresholds, a configuration-file policy engine (not an ML model), and a composite Evidence Reliability Score (ERS) on every evidence packet.

Sentinel-BLR processes live camera feeds under real-world environmental stress — night, rain, motion blur, monsoon fog, sodium vapour streetlighting — and detects all seven categories of traffic violation defined in the problem statement.

Four Architectural Principles

1. Detect violations reliably under adverse Bengaluru conditions.
2. Gate enforcement on evidence quality, not raw detection confidence.
3. Keep all enforcement policy inspectable and updateable without model redeployment.
4. Produce legally defensible, tamper-evident evidence packets on every confirmed violation.

2. System at a Glance — Naive Approach vs. Sentinel-BLR

Problem	Naive Approach	Sentinel-BLR v4.2
Camera drift	Manual recalibration	Auto-Calibration Heartbeat (SIFT/ORB + H^{-1} ROI re-projection) every 15 min
Dirty / occluded plates	Single-frame OCR	Temporal Character-Level Probability Fusion across 5 frames + Levenshtein outlier filter
Dense traffic triple riding	Pose estimation (fails in crowds)	Volumetric Bounding-Box Profiling (primary at high-density junctions, crowd index > 0.35); MediaPipe Pose Lite (low-density enhancement, index ≤ 0.35); both require 3-frame consensus
False automated fines	Global confidence threshold	Per-violation thresholds + Layer 4A config policy engine + ERS; OCR confirmation required for auto-fine
Compute feasibility	Theoretical benchmark	ATS + TensorRT INT8 $\rightarrow$ 28 ms typical / 38 ms peak design target on Jetson Orin NX (includes 3–5 ms IPC/memory-transfer margin; to be measured in Phase 1)
Kafka evidence bandwidth	Raw frame streaming (prohibitive)	Hybrid Evidence Payload: JSON metadata + Base64 compressed ROI JPEG (< 50 KB) — cloud review without full-frame streaming
Safety violation data scarcity	Zero-initialise or ignore	Synthetically generated priors from public Karnataka district road statistics (KA-TRSI) + pre-trained Indian traffic weights (IDD, ACFR, Roboflow Helmet); explicitly flagged synthetic; live data dominates weight within 30 days

3. What This System Gets Right — And What It Does Not (Yet)

Strengths

Capability	Detail
Full violation coverage	All seven violation categories in a single integrated pipeline
Computationally feasible	TensorRT INT8 + ATS + trigger gating + async queues $\rightarrow$ 28 ms typical / 38 ms peak design target on Jetson Orin NX; Phase 1 p95 target ≤ 38 ms
Self-healing geometry	SIFT/ORB landmark homography heartbeat detects and re-projects spatial ROIs without halting enforcement

Capability	Detail
Robust triple-riding	Volumetric BB Profiling is the primary method (dominant at high-density Bengaluru junctions, index > 0.35); MediaPipe Pose Lite activates as a precision enhancement at low-to-medium density (index ≤ 0.35); both paths require 3-frame consensus
Temporal OCR fusion	Character-level softmax probability summed across 5 frames; Levenshtein outlier filtering; argmax per position yields final plate string
Repeat offender tracking	Cross-junction plate-violation index; REPEAT_OFFENDER flag surfaces MV Act 2019 §210A enhanced penalty authority
GradCAM explainability	Saliency overlay on every flagged frame feeds reviewer acceleration, legal defensibility, and active learning fine-tuning loop
Evidence Reliability Score	Single composite 0–100 ERS on every packet; one number replaces five fields for officers and courts
Policy engine as config	Enforcement rules updated without redeploying any model; changes take effect in under a minute
Hybrid Kafka payload	JSON metadata + compressed ROI JPEG (<50 KB) enables cloud human review without raw-frame streaming bandwidth

Remaining Limitations and Mitigations

Limitation	Current Mitigation	Future Path
Kannada-script plates	Temporal probability fusion + LOW_CONFIDENCE flag routes to human review	Fine-tune PaddleOCR on Kannada-script plate dataset (Phase 2)
Night-time triple riding (IR-only)	IR-mode flag triggers luminance-based head-count heuristic; routed to human review	Dedicated IR-trained classifier
Synthetic prior accuracy	Synthetically generated from public Karnataka district road statistics; not observed enforcement records; explicitly flagged as synthetic in all dashboard outputs; downweighted by live data (<20% weight after 30 days)	Replace entirely with local Phase 1 data within 30 days
Single-camera blind spots	Deployment panel flags junctions needing multi-angle coverage based on OSM geometry	Phase 3 multi-camera rollout

4. Deployment Pathway

Phase	Scope	Gate Criterion
Phase 1	10 high-urgency junctions (Dataset B top-10 parking urgency)	Go/no-go: Layer 2 mAP@0.5 $\geq$ 0.85 AND Layer 4 Precision $\geq$ 95% on Phase 1 data
Phase 2	50 junctions, selected by live Sentinel-BLR urgency scores	Synthetic priors replaced by locally grounded live data within 30 days of Phase 1
Phase 3	City-wide rollout with multi-angle coverage recommendations from Layer 9	Infrastructure cost grows linearly or sub-linearly with junction count

5. Problem Statement and Gap Analysis

Why Manual Enforcement Fails

Bengaluru’s road network spans approximately 14,000 km. The BTP’s CCTV network generates far more footage than any human team can review in real time. Manual enforcement has three structural failure modes:

- **Scale** — more camera-hours than human officer capacity.
- **Environmental** — degraded performance in low-light, rain, and haze.
- **Strategic** — deployment based on officer intuition rather than violation density evidence.

Sentinel-BLR closes all three.

What Earlier Versions Did Not Get Right

Critique (v3.0 Red-Team)	v4.2 Structural Response
Computational budget: full pipeline at 25 fps is infeasible	Trigger-based inference + TensorRT INT8 + ATS + async queues → 28 ms typical / 38 ms peak design target (3–5 ms IPC margin accounted for). Phase 1 p95 validation target $\leq$ 38 ms. See Layer 2A.
MediaPipe Pose in dense traffic produces limb-attribution errors; primary path inadequate at high crowd density	Architecture inverted: Volumetric BB Profiling is now the primary path at high-density junctions (crowd index $>$ 0.35, the dominant condition at Bengaluru peak-hour junctions); MediaPipe Pose Lite activates as a precision enhancement at low-to-medium density (index $\leq$ 0.35). Mismatches between paths route to human review.
OCR defeated by mud, damage, and Kannada script	Temporal Character-Level Probability Fusion across 5 frames; softmax sum + argmax; Levenshtein outlier filter

Critique (v3.0 Red-Team)	v4.2 Structural Response
Calibration drift when cameras shift	Auto-Calibration Heartbeat via SIFT/ORB landmark homography every 15 min; automatic ROI re-projection
Dataset B has near-zero safety violation examples	Synthetic priors generated from public Karnataka district road statistics (KA-TRSI; explicitly flagged in all outputs) + pre-trained Indian traffic weights (IDD, ACFR, Roboflow Helmet); live data dominates within 30 days
Kafka raw frame streaming is network-prohibitive	Hybrid Evidence Payload: Kafka carries JSON metadata + Base64 ROI JPEG crop (<50 KB). Never raw video.

6. System Uniqueness and Competitive Differentiation

KEY DIFFERENTIATOR: Adaptive Temporal Striding (ATS)

In adverse weather, rather than suppressing sub-models (which reduces detection coverage) or running every sub-model (which blows the budget), Layer 2A applies ATS: classification runs on every 5th frame for affected tracks. A vehicle remains in the ROI for ~ 30 –60 frames, yielding 6–12 classification samples — sufficient for high-confidence multi-frame consensus while staying within the 40 ms per-frame budget. ATS replaces suppression as the graceful degradation strategy.

KEY DIFFERENTIATOR: Computationally Honest at Every Layer

v4.2 is the first version of Sentinel-BLR that can be honestly presented as real-time on real hardware. The combination of TensorRT INT8, trigger-gated sub-model activation, per-Track-ID inference gating, ATS, and async queues produces a design-target cycle of 28 ms typical / 38 ms peak on Jetson Orin NX — within the 40 ms budget for 25 fps. End-to-end $p95 \leq 38$ ms is the Phase 1 measured validation target; the design target is not presented as a measured benchmark.

KEY DIFFERENTIATOR: Self-Healing Spatial Geometry

Auto-Calibration Heartbeat is absent from virtually every ANPR or CV traffic enforcement system currently deployed in India. Camera drift is not hypothetical; it is routine. Without auto-calibration, a virtual stop-line polygon can drift 1–3 metres of projected ground distance per degree of camera tilt — silently producing false positives or missed violations. Sentinel-BLR detects drift within 15 minutes, re-projects all spatial ROIs automatically, and notifies operators without halting enforcement.

KEY DIFFERENTIATOR: Temporal Character-Level Probability Fusion

Standard multi-frame OCR voting takes the highest-confidence full string across frames. Sentinel-BLR's approach is fundamentally different: it stores softmax probability vectors for each character position independently across 5 frames, sums the probability matrices, and applies argmax per character position. This means a character that is clearly visible in 2 frames but occluded in 3 still contributes its probability mass — allowing partial-visibility frames to contribute to a complete plate read. Levenshtein distance is used only as an outlier filter to discard corrupt frames before fusion, not as the voting mechanism.

KEY DIFFERENTIATOR: Hybrid Evidence Payloads over Kafka

The architectural tension: human reviewers need to see the violation frame, but streaming raw video over Kafka at multi-junction scale is network-prohibitive. Sentinel-BLR resolves this with Hybrid Evidence Payloads: each Kafka message carries full JSON metadata plus a Base64-encoded compressed JPEG crop of the violation ROI ($\sim 400 \times 400$ pixels, < 50 KB). Reviewers see the violation without full-frame streaming.

KEY DIFFERENTIATOR: Policy Engine as Configuration, Not ML

The enforcement policy engine (Layer 4A) is a YAML configuration file. BTP can update enforcement rules — add a diversion polygon, raise the OCR confidence floor, extend a warning-only phase for a new junction, suspend auto-fining for a violation type under legal review — without redeploying any model. Changes take effect in under a minute. The enforcement decision-making logic is inspectable, auditable, and modifiable by non-engineers.

KEY DIFFERENTIATOR: Evidence Reliability Score (ERS)

Every evidence packet carries a single composite integer (0–100) distilling detection confidence, OCR quality, frame consensus depth, calibration health, and preprocessing quality into one number that officers, reviewers, and courts read at a glance. The ERS formula is:

$$\text{ERS} = (0.30 \times \text{detection_confidence}) + (0.25 \times \text{OCR_confidence}) + (0.20 \times \text{frame_consensus_ratio}) + (0.15 \times \text{calibration_health}) \quad (1)$$

No new model is required; all components are already computed upstream.

KEY DIFFERENTIATOR: Closed-Loop Strategic Intelligence

Violations detected today update the junction heatmap tonight, which updates deployment recommendations tomorrow. The recency-decay formula ($0.95^{\text{days_ago}}$) rapidly shifts weight from historical baselines to live data. This self-improving cycle is not present in documented single-violation ANPR deployments in Indian cities.

7. System Architecture — Full Pipeline v4.2

The pipeline processes camera feeds through fifteen distinct layers, each with a defined responsibility and explicit computational budget. Detection and enforcement are deliberately separated: Layer 3 detects; Layer 4A applies enforcement policy as a configuration file, not an ML model.

Three Evidence-Quality Layers — Layer 1A (auto-calibration), Layer 2A (inference budget controller + ATS), and Layer 4A (enforcement policy engine) — together constitute the architecture's spine.

Pipeline Architecture Diagram

Sentinel-BLR :- System Architecture



Layer Reference Table

Layer	Function	Technical Detail	Group
0	Feed Ingestion	RTSP/ONVIF camera streams; edge-local ring buffer; Kafka Hybrid Evidence Payloads (JSON + ROI JPEG, never raw video); dynamic 5–30 fps per junction	INGESTION
1	Environmental Preprocessing	4-stage GPU-accelerated pipeline (CLAHE, Multi-Scale Retinex, DCP de-hazing, Wiener de-blur); conditionally activated per frame	PREPROC
1A	Auto-Calibration Heartbeat	SIFT/ORB landmark extraction every 15 min; homography H between reference and current; drift alert > 1.2 px; inverse H^{-1} re-projects all spatial ROIs automatically	PREPROC
2	Macro Detection	YOLOv10-N TensorRT INT8 at 640px ($\sim 4\text{--}7$ ms); ByteTrack edge-local; 11 classes (10 vehicle + unified person)	DETECTION
2A	Inference Budget Controller + ATS	Trigger-gated sub-model activation; async priority queues; per-Track-ID gating (0.5s window); Adaptive Temporal Striding (every 5th frame in adverse weather)	DETECTION
3	Violation Detection Engine	5 parallel triggered paths: (A) Helmet/Seatbelt, (B) Triple Riding hybrid, (C) Wrong-Side, (D) Stop-Line/Red-Light, (E) Illegal Parking	CLASSIF.
4	Confidence Gate	Per-violation-type multi-frame consensus thresholds; three-way escalation router: AUTO-FINE / HUMAN REVIEW / DISCARD	DECISION
4A	Enforcement Policy Engine	YAML config file (not ML); BTP-updatable in under 1 min; graduated warning tier for initial deployment; diversion polygon support; OCR-gate and calibration-gate enforcement rules	DECISION
5	License Plate OCR	PaddleOCR PP-OCRv4; triggered only post-Layer 4; Temporal Character-Level Probability Fusion (5-frame softmax sum + argmax); Levenshtein outlier filter; KA-regex gate; 0/O, 1/I, 8/B, S/5 correction	ID

Layer	Function	Technical Detail	Group
5B	Repeat Offender Tracking	Cross-junction plate-violation index (TimescaleDB); 24h/7d/30d lookback windows; REPEAT_OFFENDER flag; MV Act 2019 §210A statutory grounding; HIGH_CONFIDENCE writes only	ID
6A	GradCAM Explainability	Saliency overlay on all flagged frames; async low-priority queue (no pipeline latency); feeds reviewer acceleration, legal defensibility, and active learning fine-tuning	EVIDENCE
6	Evidence Packet Generator	Annotated frame + GradCAM overlay + UTC timestamp + GPS + junction ID + plate text + violation type + vehicle class + review metadata + SHA-256 hash + ERS (0–100)	EVIDENCE
7	Persistence	PostgreSQL + TimescaleDB (violation records, heatmap stats); DuckDB (batch analytics on Dataset B); nightly 02:00 aggregation	STORAGE
8	Strategic Intelligence	Junction Urgency Score $= \sum [\text{severity_weight} \times \text{hourly_count}] \times 0.95^{\text{days}}$; synthetic priors <20% weight after 30 days; weekly LoRA fine-tuning loop	INTEL
9	Analytics Dashboard	8 panels: Live Violation Feed Map, Junction Heatmap, Peak-Hour Trends, Human Review Queue, Deployment Recommendations, Model Performance Tracker, Calibration Health, Public Compliance Feed (anonymised)	OUTPUT

8. Layer-by-Layer Technical Specification

The Compute Story: How 40 ms Is Met

The four preprocessing stages in Layer 1 are not all active simultaneously. They are conditionally gated:

Stage	Activation Condition	Typical Latency
CLAHE (contrast equalisation)	Always active	~5 ms

Stage	Activation Condition	Typical Latency
Multi-Scale Retinex (colour balance)	Always active	~8 ms
Dark Channel Prior de-hazing	Activated by IMD API rain/fog index OR Dataset A waterlogging pre-schedule	+12 ms additional
Wiener Deconvolution de-blur	Activated when Laplacian variance pre-check detects motion blur	+10 ms additional

Full pipeline budget (design target) — Jetson Orin NX:

Stage	Typical	Peak (adverse wx.)	Extreme (all 4 stages)
Preprocessing (conditional)	13 ms	25 ms	35 ms
YOLOv10-N TensorRT INT8	4 ms	7 ms	7 ms
ByteTrack (edge-local)	2 ms	4 ms	4 ms
MobileNetV3 (when triggered)	3 ms	6 ms	ATS stride
Triple-riding module (when triggered)	8 ms	ATS stride	ATS stride
End-to-end frame cycle	28 ms	38 ms	38 ms (ATS)

Under extreme conditions (all 4 preprocessing stages active), Layer 2A applies Adaptive Temporal Striding: classification sub-models process every 5th frame for tracked vehicles rather than every frame. ATS replaces sub-model suppression as the graceful degradation strategy, maintaining detection coverage while managing compute.

Layer 1A — Auto-Calibration Heartbeat

Every 15 minutes, the Auto-Calibration Heartbeat runs as a background process independent of the main inference pipeline. Without auto-calibration, a virtual stop-line polygon can drift 1–3 metres of projected ground distance per degree of camera tilt — enough to falsely flag compliant vehicles or miss genuine violations.

Mathematical grounding:

- **Installation:** A Calibration Reference Frame is generated — 40–60 stable visual landmarks (kerb edges, road markings, signal poles, wall corners) extracted via SIFT/ORB feature detection, stored with projected ground-plane coordinates.
- **Every 15 min:** Heartbeat extracts the same landmark set from the live frame and computes homography matrix H between current and reference positions.
- **Drift threshold:** If H shows translation or rotation > 1.2 px at 1080p ($\sim 0.08^\circ$ tilt), a Calibration Drift Alert is raised.

- **On alert:** All spatial ROIs (stop-line polygons, no-parking zones, Safe-Flow Vectors) are automatically re-projected using inverse homography: $\mathbf{p}' = H^{-1}\mathbf{p}$. Enforcement continues without interruption.
- **Degraded mode:** If landmark match rate falls below 50%, the junction is flagged **Calibration-Uncertain** and all detections route to human review.
- **Guard condition:** Scene occupancy monitor delays heartbeat when heavy vehicle coverage > 50% of frame; retries within 60 seconds. Maximum drift-window cap: if calibration cannot be obtained within 45 minutes of the scheduled heartbeat, the junction enters **Calibration-Uncertain** mode.

Layer 3 — Violation Detection Engine: Five Parallel Paths

- **Path A — Helmet / Seatbelt:** MobileNetV3 binary classifier (triggered only on two-wheeler or car bounding box from Layer 2). Initialized from IDD + ACFR + Roboflow Helmet pre-trained weights. 3-frame consensus required before escalation to Layer 4.
- **Path B — Triple Riding (Hybrid Module):**
 - *Primary method (high-density, crowd index > 0.35):* Volumetric Bounding-Box Profiling — aspect ratio and pixel density of the motorcycle bounding box are analysed; flag raised if vertical/horizontal mass distribution exceeds $\Delta \geq 1.2\times$ single-rider baseline. 3-frame consensus required.
 - *Precision-enhancement method (low-to-medium density, crowd index ≤ 0.35):* Chassis-Person Centroid Association with velocity coupling. MediaPipe Pose Lite extracts torso centroids within the expanded two-wheeler bounding box. Only centroids whose velocity vector is within 15° of vehicle heading and 0.5 m/s magnitude tolerance are counted as riders. 3-frame consensus required.
 - Disagreement between both active paths routes to human review.
- **Path C — Wrong-Side Driving:** Safe-Flow Vector maps generated at calibration via SIFT/ORB auto-estimation from OSM road geometry. Deviation angle threshold: 25° default, adjustable per junction. Two-frame sustain requirement. Minimum velocity filter eliminates stationary U-turns.
- **Path D — Stop-Line and Red-Light Crossing:** Virtual stop-line polygons defined at calibration. Signal state determined by a YOLO lamp-colour classifier (primary operative method — trained to classify signal head crop as RED/AMBER/GREEN, robust to sodium-vapour lighting and rain). BTP Traffic Signal API integration is a Phase 2 aspirational enhancement. Violation fires on simultaneous centroid-in-ROI and classifier-RED conditions.
- **Path E — Illegal Parking:** Stationary-threshold filter with 180-second duration (configurable per zone type). No-parking ROI sources: manual annotation, BMTC GTFS bus-stop polygons, Dataset B parking-concentration zones. Velocity floor: 0.05 m/s to exclude red-light stops.

Layer 4 — Confidence Gate and Escalation Router

Violation Type	Auto-Flag Threshold	Human Review Band	Discard Below
Helmet / seatbelt	$\geq 88\%$ / 3 frames	60–87%	< 60%

Violation Type	Auto-Flag Threshold	Human Review Band	Discard Below
Triple riding	$\geq 92\%$ / 3 frames	65–91%	$< 65\%$
Wrong-side driving	$\geq 87\%$ / 2 frames	60–86%	$< 60\%$
Stop-line / red-light	$\geq 90\%$ / 3 frames	60–89%	$< 60\%$
Illegal parking	$\geq 85\%$ / sustained	55–84%	$< 55\%$

Triple riding receives the highest auto-flag threshold (92%) because density-related false-positive risk is elevated in dense markets. Illegal parking receives the lowest (85%) because sustained duration provides structurally stronger evidence.

Layer 4A — Enforcement Policy Engine

Violation	Condition	Enforcement Action
Triple riding	Any confidence	Human approval required — never auto-fine
Helmet / seatbelt	$\geq 88\%$, 3 frames, OCR confirmed	Auto-fine
Helmet / seatbelt	$\geq 88\%$, 3 frames, OCR low-confidence	Human approval required
Wrong-side driving	Near active diversion polygon	Human review only
Stop-line / red-light	$\geq 90\%$, 3 frames, signal confirmed	Auto-fine
Illegal parking (1st offence)	$\geq 85\%$, sustained 180–300 s, OCR confirmed, no prior in 30-day window	Warning only (civic trust phase; auto-fine on 2nd offence within 30 days)
Any violation	OCR confidence $< 90\%$	Human approval — never auto-fine
Any violation	Calibration-Uncertain junction	Human review only — no auto-fine

Layer 5 — License Plate OCR with Temporal Probability Fusion

Standard multi-frame OCR voting selects the highest-confidence full string across frames. Sentinel-BLR’s approach is fundamentally different:

- **Per-character softmax storage:** Softmax probability vectors for each character position are stored independently across up to 5 frames, chosen for maximum plate visibility (bounding box aspect ratio).

- **Matrix summation:** Probability matrices are summed element-wise across all frames. Characters clearly visible in 2 frames but partially occluded in 3 still contribute their probability mass.
- **Argmax per position:** Final plate string is the argmax of the summed matrix per character position — yielding the highest-probability character at each position across all frames.
- **Levenshtein outlier filter:** Applied before fusion to discard corrupt frames (e.g. mud splash obscuring all characters). Edit distance threshold filters outliers; this is not the voting mechanism.
- **Majority-vote confirmation:** If majority string exists (at least 3/5 frames agree within edit distance 1), that string is the confirmed plate read. If not, LOW_CONFIDENCE flag routes to human review.

Levenshtein correction pairs for Indian number-plate font: 0/O, 1/I, 8/B, S/5. KA-regex gate validates Karnataka plate format before indexing.

Layer 6 — Evidence Packet + Evidence Reliability Score (ERS)

Each confirmed violation generates a tamper-evident evidence packet containing: annotated frame with GradCAM overlay, UTC timestamp, GPS coordinates, junction identifier, plate text with confidence, violation type and offence code, vehicle class, review metadata, SHA-256 hash of annotated frame bytes, and ERS (0–100).

$$\text{ERS} = (0.30 \times \text{detection_confidence}) + (0.25 \times \text{OCR_confidence}) + (0.20 \times \text{frame_consensus_ratio}) + (0.15 \times \text{calibration}) \quad (2)$$

ERS routing: ≥ 85 — auto-fine eligible; 60–84 — human review queue; < 60 — discarded. The SHA-256 hash makes any subsequent modification to the annotated frame detectable in legal proceedings.

Active Learning Feedback Loop

- **Label generation:** Every human accept/reject in the 60–92% confidence band is recorded as a labelled training example (image crop, ground-truth label, reviewer ID, timestamp) in a dedicated fine-tuning queue.
- **Weekly LoRA fine-tuning:** Sunday 02:00 maintenance window applies Low-Rank Adaptation (LoRA) updates to MobileNetV3. LoRA restricts parameter updates to low-rank decomposition matrices, preventing catastrophic forgetting while adapting to local Bengaluru conditions at minimal compute cost.
- **Model drift detection:** Rolling reviewer-override rate on the most recent 500 labelled examples is monitored in Layer 9 dashboard. Rising override rate triggers ahead-of-schedule fine-tuning.

9. Operational Safeguards

Risk	Mitigation	Layer
Triple-riding false positives	Human approval always required regardless of confidence level	4A

Risk	Mitigation	Layer
OCR failure on dirty/Kannada plates	No OCR confirmation = no auto-fine; routes to human review automatically	4A + 5
Camera occlusion during calibration	Scene occupancy monitor delays heartbeat if vehicle coverage > 50%; retries in 60 s; 45-min max drift-window cap → Calibration-Uncertain mode if calibration cannot complete	1A
Wrong-side context staleness	Officer draws diversion polygon via dashboard with expiry timestamp; 4A enforces no auto-fine within polygon	4A + 9
Review queue overload	Adaptive confidence floor: threshold rises when queue depth exceeds configurable limit, reducing inflow until queue clears	4
Privacy (non-violating road users)	Faces of non-violating users blurred before storage (PDPB 2023 compliance); only violation-confirmed frames stored at full resolution; raw frames deleted within 24 hours	6 + config
Multi-camera blind spots	Deployment panel flags junctions where multi-angle coverage would materially reduce blind spots (OSM geometry)	9 + roadmap
Plate misread + repeat-offender index	Only HIGH_CONFIDENCE reads (OCR $\geq 90\%$, majority-vote consensus) are indexed; LOW_CONFIDENCE reads never written to repeat-offender index	5B

10. Data Integration Strategy

Source	Records	Used For	Explicitly NOT Used For
Dataset B	298,449	Parking spatial heatmap; peak-hour patterns by vehicle class; junction deployment priority ranking	Safety violation urgency scoring (2 helmet records in 298k is not a dataset for safety modelling)
Dataset A	8,204	Waterlogging events (458): pre-schedule DCP de-hazing. Accident events (365): 1.15× urgency multiplier. Pothole events: calibration notes for lane-weaving	—

Source	Records	Used For	Explicitly NOT Used For
Synthetic priors (KA-TRSI; generated from public Karnataka district road statistics; explicitly labelled synthetic in all outputs)	District-level	Safety-violation hourly count initialisation; disaggregated to junction level via Dataset B vehicle-class distributions; < 20% weight contribution after 30 days live data	Never presented as observed enforcement data
IMD Weather API	Real-time	DCP de-hazing activation (15-min polling); fallback to Dataset A waterlogging schedule	—
YOLO lamp-colour classifier (primary); BTP Signal API (Phase 2, requires BTP partnership)	Real-time	Signal state for stop-line detection	—
BMTC GTFS	Static	Bus-stop no-parking zone polygons	—
OSM / Maps Roads API	Static	Safe-Flow Vector initialisation; junction geometry for multi-angle blind-spot analysis	—

11. Technical Stack

Component	Technology	Rationale
Object detection	YOLOv10-N, TensorRT INT8	~4 ms inference; 2× faster than Efficient variant; INT8 calibrated on 500 Bengaluru traffic frames to prevent accuracy collapse on auto-rickshaws and BMTC buses
Multi-object tracking	ByteTrack (edge-local)	No raw frame transmission; Track IDs managed on-device; only metadata and ROI payloads upstream
Safety classification	MobileNetV3 (trigger-gated)	Binary classification; never runs without two-wheeler trigger; initialized from IDD + ACFR + Roboflow Helmet weights

Component	Technology	Rationale
Pose / centroid	MediaPipe Pose Lite (density-gated)	Volumetric BB Profiling is primary at high-density junctions (index > 0.35); MediaPipe Pose Lite activates as precision enhancement at low-to-medium density
Preprocessing	OpenCV 4.x, CUDA build	CLAHE, Multi-Scale Retinex, DCP, Wiener; GPU-accelerated; conditionally activated per frame
Deep learning framework	PyTorch 2.x	Training, fine-tuning, and LoRA active learning updates
License plate OCR	PaddleOCR PP-OCRv4	Temporal character-level probability fusion; 5-frame softmax sum + argmax; Levenshtein outlier filter
Calibration	SIFT/ORB + homography (OpenCV)	15-min drift detection; inverse homography H^{-1} for automatic ROI re-projection
Message broker	Apache Kafka 3.x	Hybrid Evidence Payload (JSON + ROI JPEG); partitioned by junction ID; no raw video
API layer	Python FastAPI	Async; WebSocket support for live dashboard updates
Time-series DB	PostgreSQL + TimescaleDB (PG 16 + TS 2.x)	High-write time-series; temporal SQL; repeat-offender index; nightly aggregation at 02:00
Batch analytics	DuckDB 0.10x	298k-record Dataset B queries in < 1 s; junction urgency computation
Dashboard	Streamlit + Folium + Plotly	GIS-capable; 8 operational panels; calibration health panel; public compliance feed
Containerisation	Docker + docker-compose	Full-stack deployment; reproducible environment
Edge hardware (target)	NVIDIA Jetson Orin NX 16/64 GB	100 TOPS; full TensorRT pipeline at 25 fps; outdoor surveillance form factor; ATS + trigger-gating validated against Jetson TOPS specification
Cloud fallback	AWS EC2 G5 / GCP T4	On-demand GPU offload if edge hardware unavailable; Hybrid Kafka payload reduces egress bandwidth requirement

12. Performance Evaluation Framework

Layer / Output	Metric(s)	Evaluation Method	Target
Layer 2 — detection	mAP@0.5, per-class precision/recall	Held-out test from Phase 1 footage, stratified by lighting/weather condition	mAP@0.5 $\geq$ 0.85; $\leq$ 10% mAP drop in adverse conditions
Layer 3/4 — auto-fine violations	Precision, Recall, F1 per type	Confusion matrix vs. officer-adjudicated outcomes from Phase 1 human review queue	Precision $\geq$ 95%; Recall $\geq$ 80%
Layer 3/4 — triple riding (human-review)	Recall, F1 (precision secondary)	Same protocol, scored separately	Recall $\geq$ 90% (every flag is human-reviewed; missed violations are the failure mode)
Layer 5 — OCR	Plate-level accuracy, CER	Stratified holdout: clean / dirty / Kannada plates; before and after temporal fusion	$\geq$ 95% clean; $\geq$ 80% degraded (post-fusion)
Layer 2A — compute latency	End-to-end p50/p95 latency	24-hour sustained load test on Jetson Orin NX at pilot junction	p95 $\leq$ 38 ms sustained (design target; Phase 1 measured validation required; 3–5 ms IPC margin included in budget)
System-wide — scalability	Junctions per edge node; Kafka + TimescaleDB throughput	Simulated load scaling: 10 $\rightarrow$ 50 $\rightarrow$ city-wide	Linear or sub-linear infrastructure cost growth

Precision and recall targets are deliberately asymmetric. Violations reaching auto-fine outcomes are held to a high precision bar (false positive = wrongly issued fine). Triple riding — always routed to human review by Layer 4A — is held to a high recall bar (the only failure mode is missing a real violation). The evaluation framework validates the policy choices already made in Layer 4A.

13. Conclusion

Sentinel-BLR is an evidence-quality system designed to survive contact with the real world — the real Bengaluru, with cameras that shift in the wind, monsoon rain that obscures plates, markets where three bikes ride side by side at two metres per second, and an enforcement environment where an automated fine issued on low-reliability evidence is a legal liability, not a civic service.

The architectural choices documented here — Adaptive Temporal Striding, crowd-density-gated Volumetric Bounding-Box Profiling, Temporal Character-Level Probability Fusion, Hybrid Evidence Payloads, auto-calibration heartbeat, confidence-separated enforcement policy, legal grounding in MV Act 2019, and a graduated enforcement tier that builds public trust before applying fines — each exist because a specific, credible failure mode demanded them.

The result is a system that is technically proficient, operationally honest, legally defensible, and in-

crementally deployable — starting at ten high-urgency junctions identified from Dataset B analysis, improving every night it runs, and scaling to city-wide deployment without architectural replacement.

— *END OF SUBMISSION* —

Sentinel-BLR v4.2 — Evidence-Quality Traffic Enforcement Framework